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ECE115 - Module5

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Google Drive folder containing files:

<https://drive.google.com/drive/folders/1A8mDtg7jwpapd6mcWQVZRQ9MVMisnB0y?usp=sharing>

5.2.1 Add the proposed functionality to the pinball machine

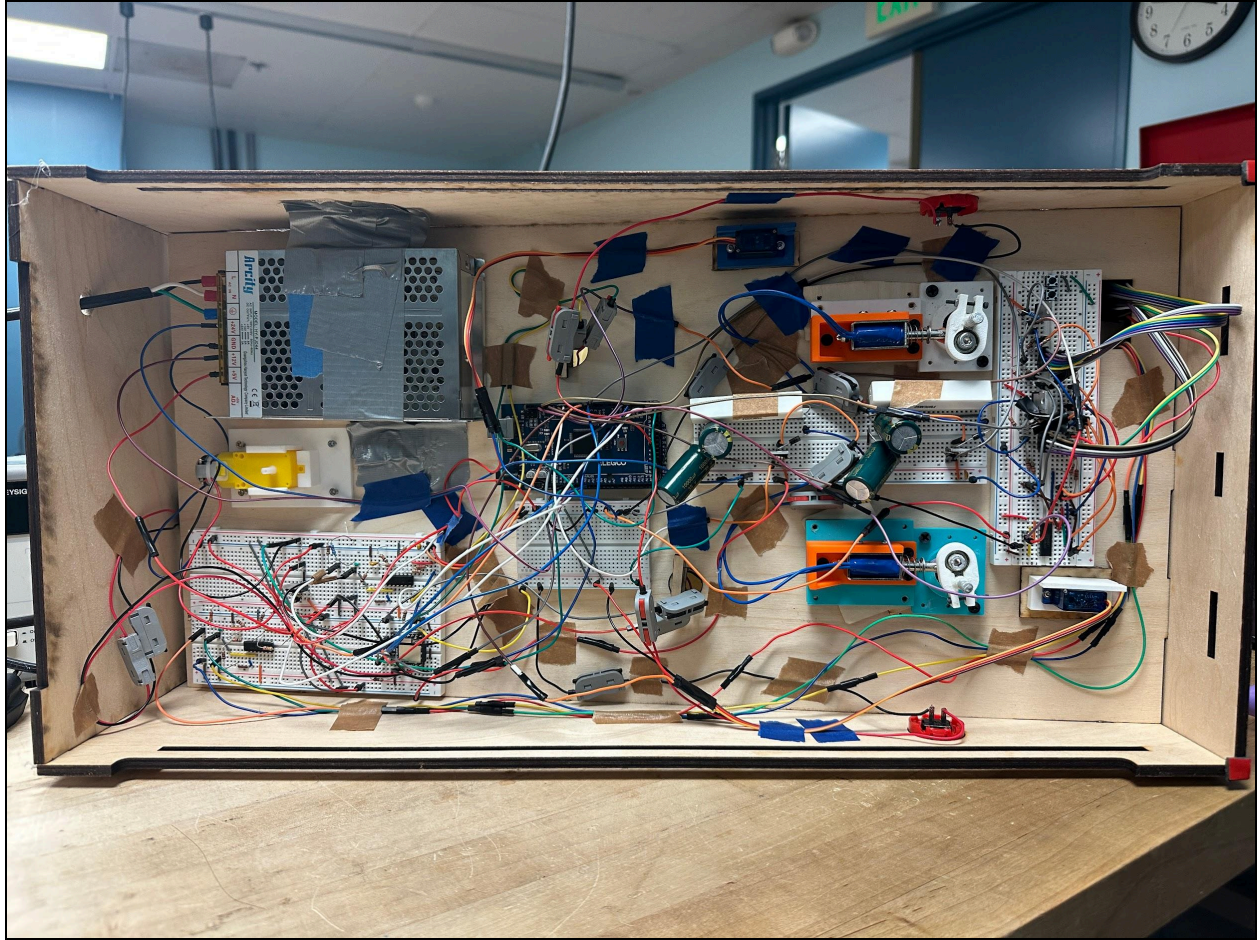
Completed additional function

Added functions: more obstacles, decals*, piezo circuitry, gate mechanism, more 3D printed figures, and compacted wire system

***reference final pinball machine photos**







Additional Functionality Omitted:

- 3D mounts for the arduino and breadboards: required longer wires for routing and created the challenge of switching between working on the top (playfield) and circuits. Switches approach by containing electronics by attaching them to the opposite of the playfield and sides of the pinball machine.
- Paint: Would take too long for paint to dry so we opted to use other techniques to improve the visuals of the pinball machine (more 3D printed figures and decals)

5.2.2 Final Presentation / Demo

Full System Evaluation: **Complete**

5.2.4 Final Writeup

Final Writeup

1. Summarize your additional functionality

Additional Functionality:

We originally proposed the added functionality of 3D printed breadboard and motor holders. However, we found that it was easier to mount the electronic components to the opposite side of the playfield and the sides, as this kept the electronics accessible when working on the top and shortened the wire lengths. One additional functionality we implemented was adding extra walls to prevent the ball from easily reaching the losing region. Furthermore, we were able to integrate piezo sensors into our system and develop an RC servo arm to load the ball into the plunger. In our proposal, we also planned to paint the pinball machine, but due to time constraints we were not able to complete this. Instead, we 3D printed SpongeBob themed figures and applied decals to make the system more visually appealing.

Requirements:

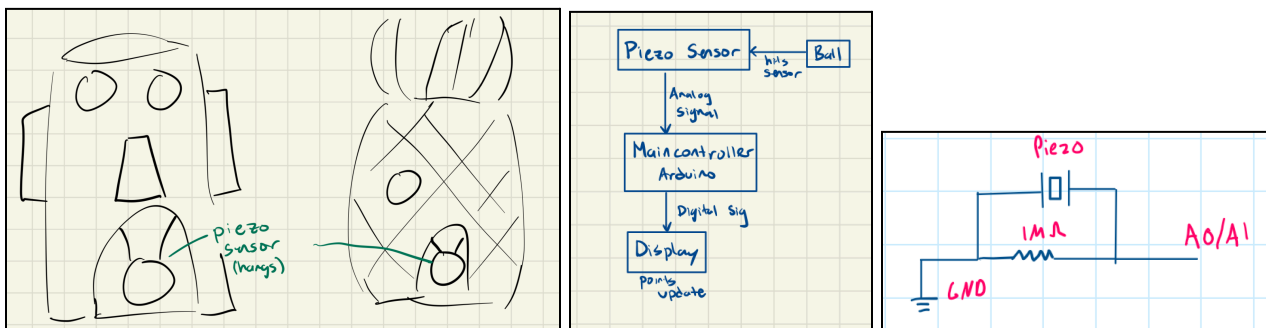
1. Flipper obstacles make losing harder
2. Piezo Sensor sensing ball well
3. Automatic gating
4. Decal doesn't obstruct playfield

Materials:

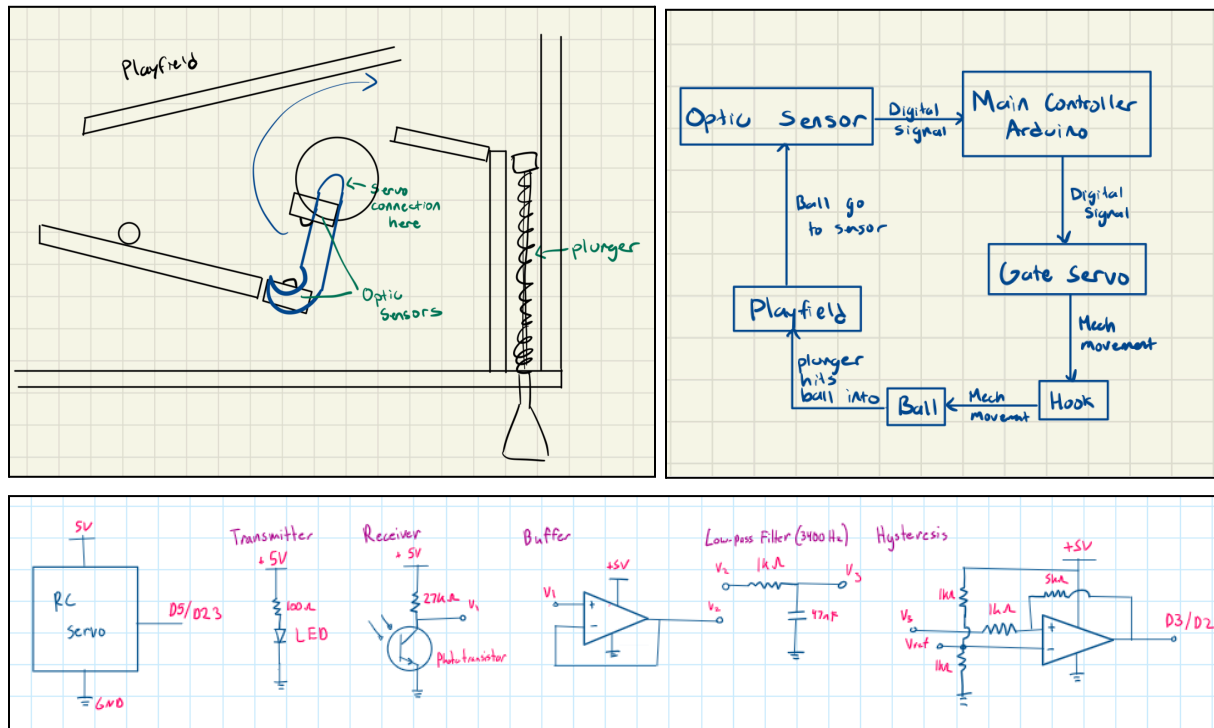
- 0.275" plywood: [Amazon](#)
- Decals: [Amazon](#)
- 3D and laser prints: Free from ECE Makerspace
 - Model files:
<https://drive.google.com/drive/folders/1kOY6Z7JHsXjdMh7Nvi-niw0s6ZCII8pl?usp=sharing>

Design:

Piezo



Gating Mechanism



Trace Matrix:

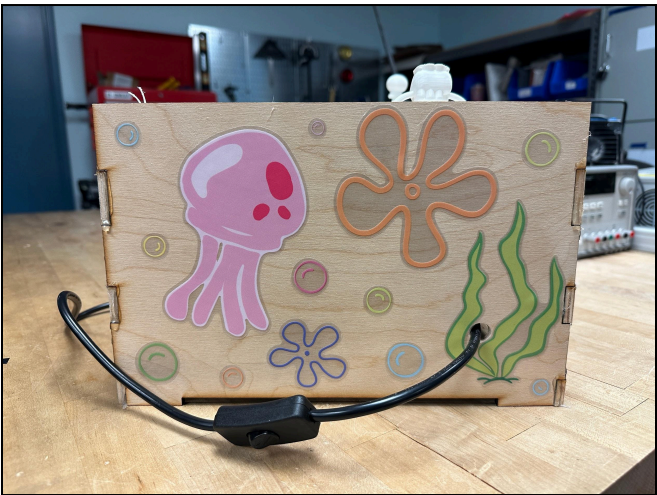
Specification number	Specification description	Test to perform	Relevant requirement	Specification (units)	Measured value (units)
1	Functionality	Launch the ball using the flipper into the piezo sensors, see if score updates.	2	Yes/No (Y/N)	Y, the squidward sensor seems to be less sensitive than spongebob and requires more force for it to update the display.
2	Functionality	Let the ball fall towards the bottom of the board and see if the gating mechanism will pick up the ball and load it in the launcher.	3	Yes/No (Y/N)	Y
3	Functionality	Let the gating mechanism load the ball 3 times, and let the ball fall towards the gating mechanism again, and see when the gating mechanism stops.	3	count	3

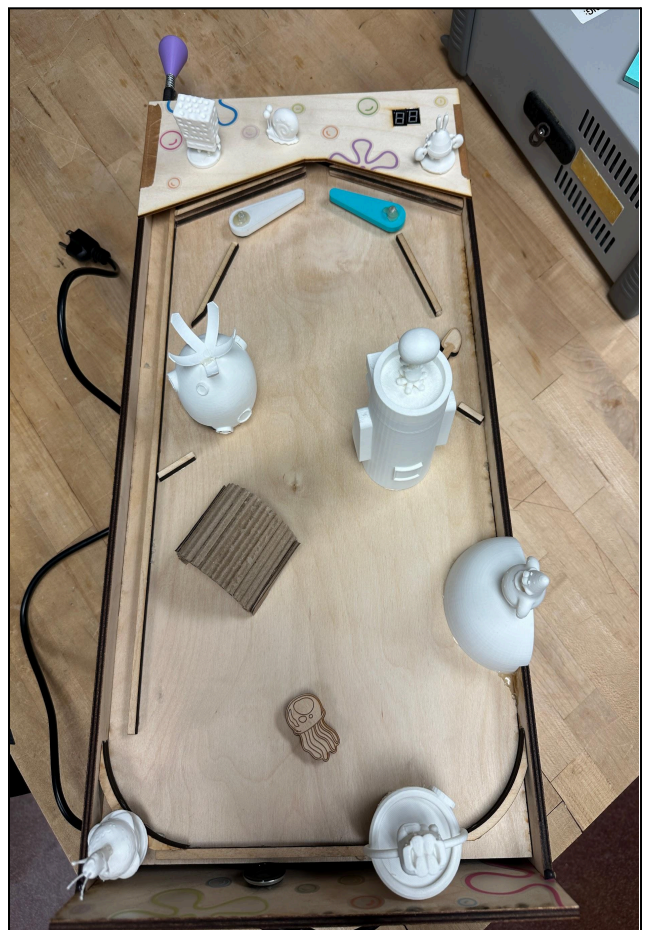
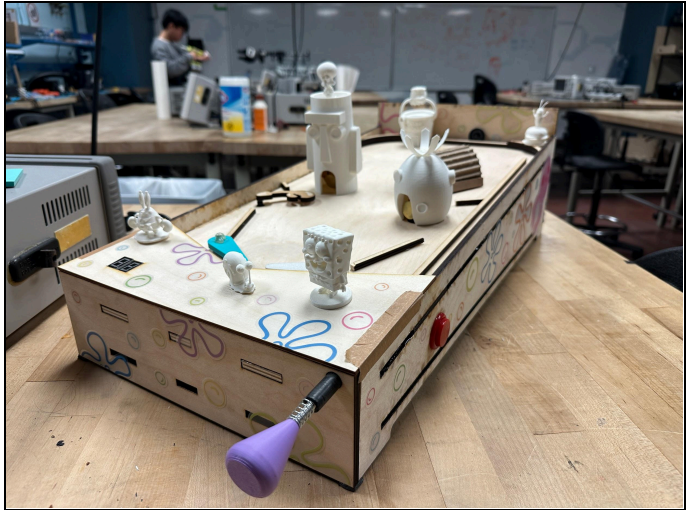
4	Functionality	Play rounds and see if they last longer with added obstacles	1	Yes/No (Y/N)	Y, rounds naturally lasted longer as they are less opportunities to instantly lose the round
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2. Summarize pinball machine

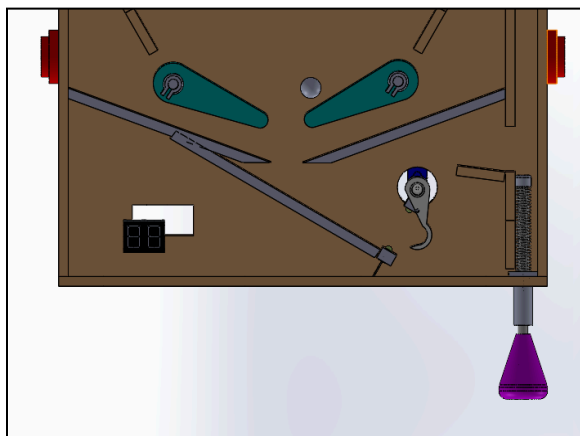
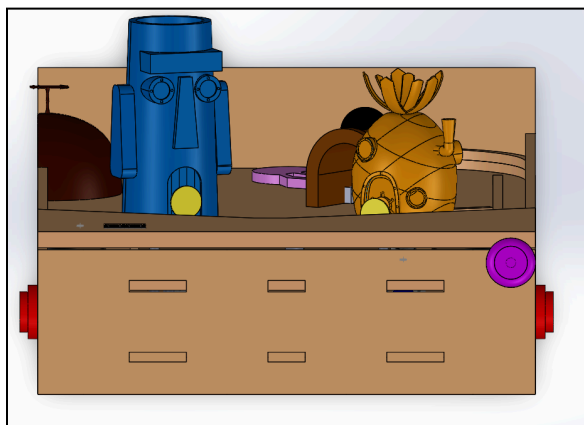
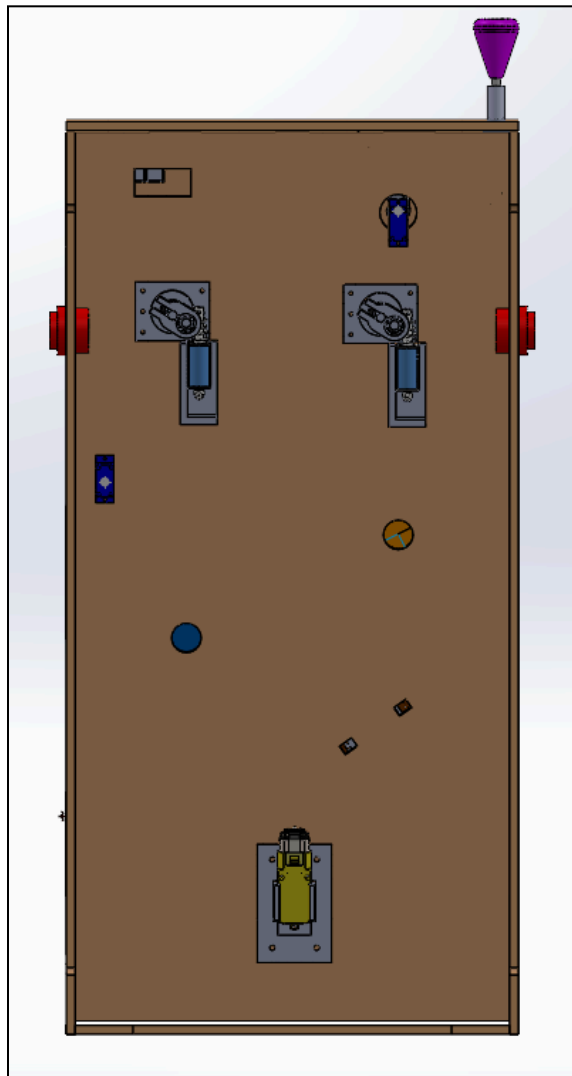
Overall Pinball Machine:

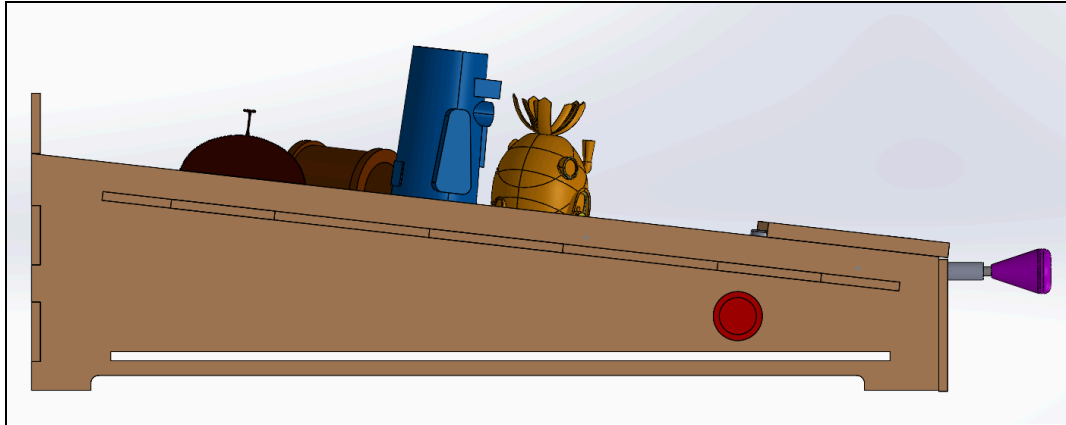
Pinball machine in its complete glory!:





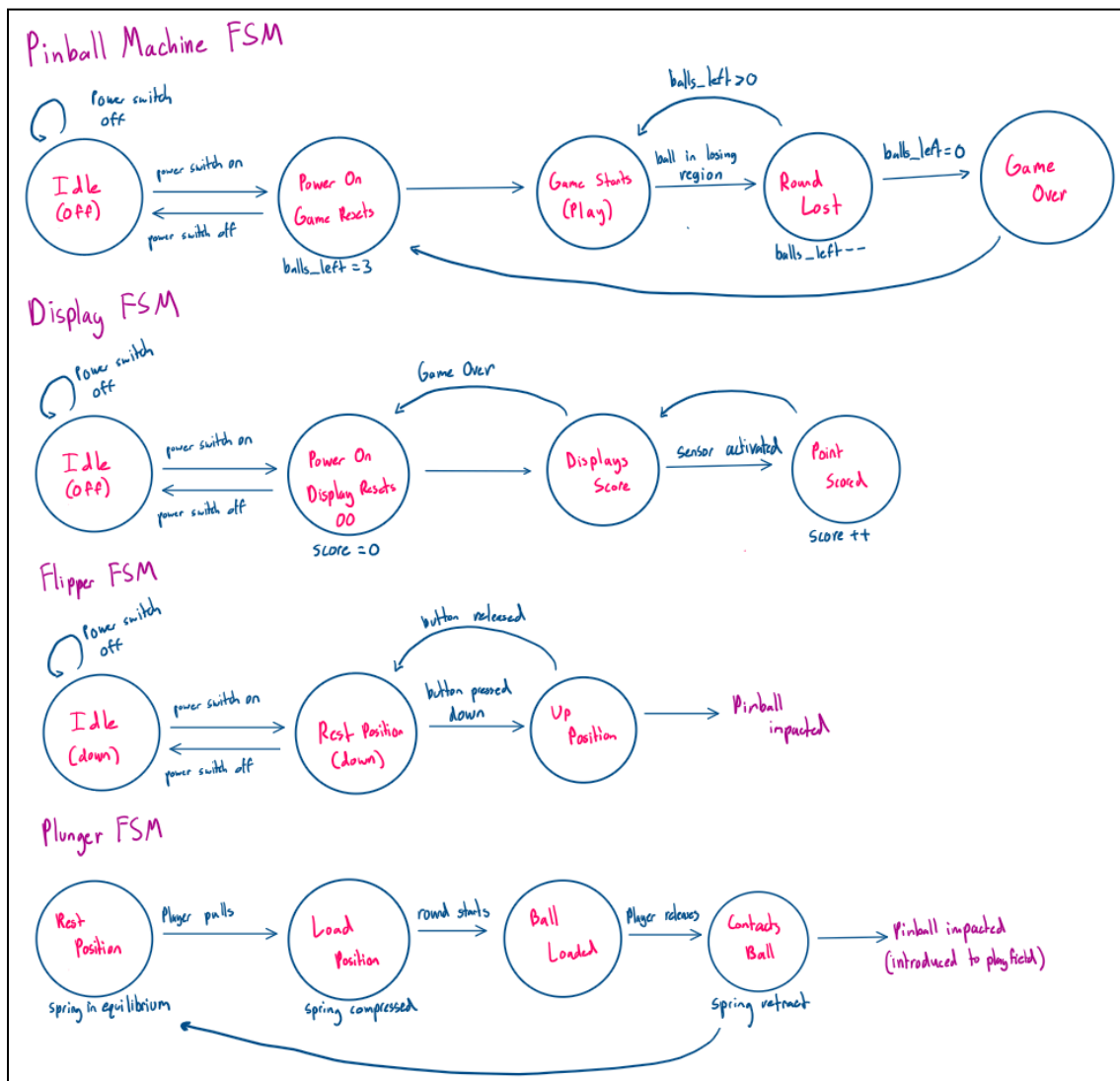
CAD:

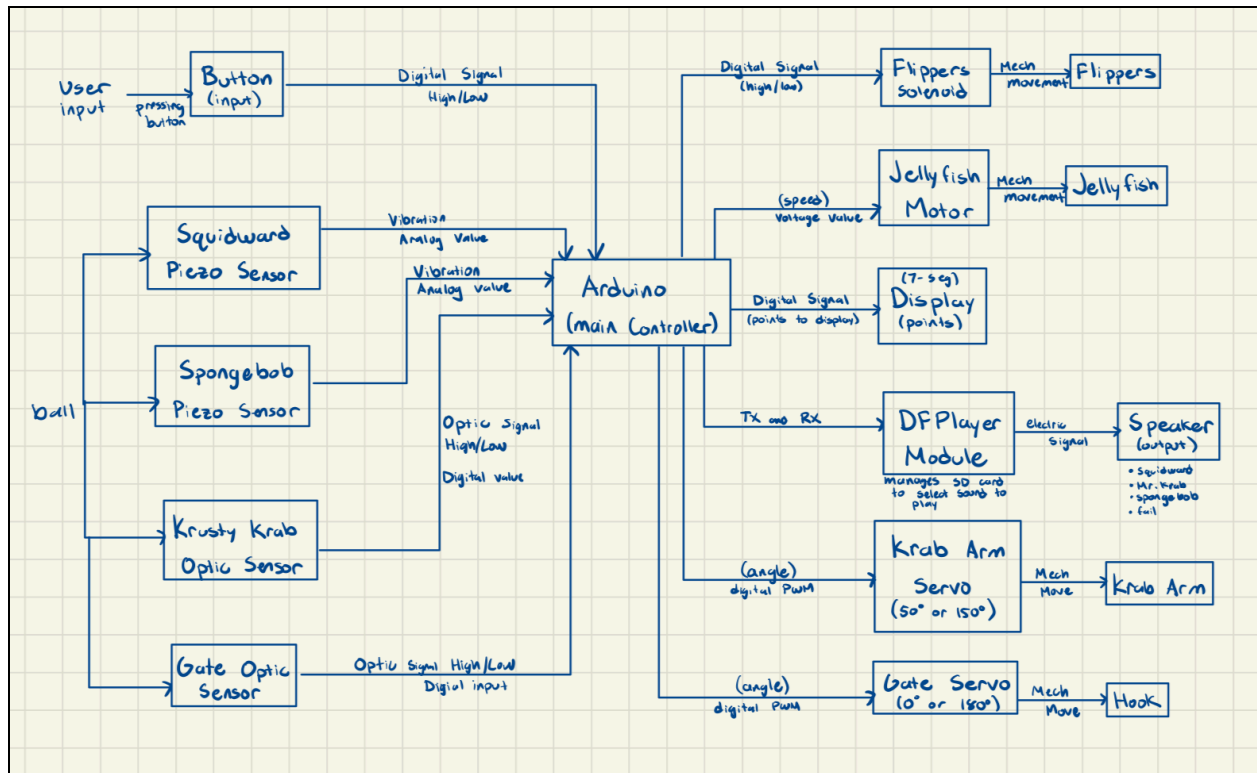




Extra decoration and figures not in CAD images. Couldn't find a Solidworks compatible file for 3D prints.

FSM and Block Diagram:

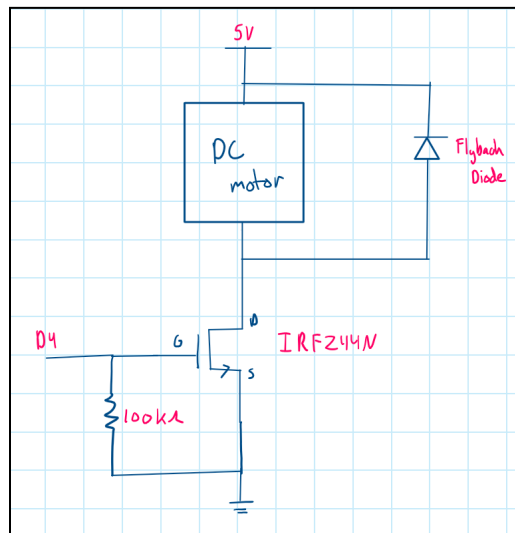




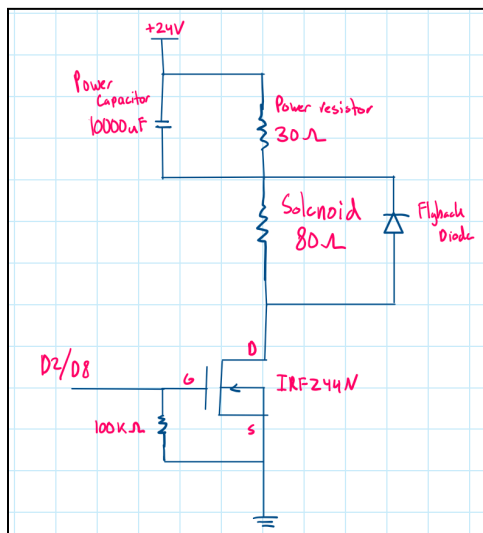
Circuit Diagrams (including Arduino connections):

Note: Both 5V and 24V power rails are decoupled with a 0.1uF capacitor. For simplicity, not drawing in each diagram as it would be the same component for every system. Arduino was powered through the 12V output of the power supply.

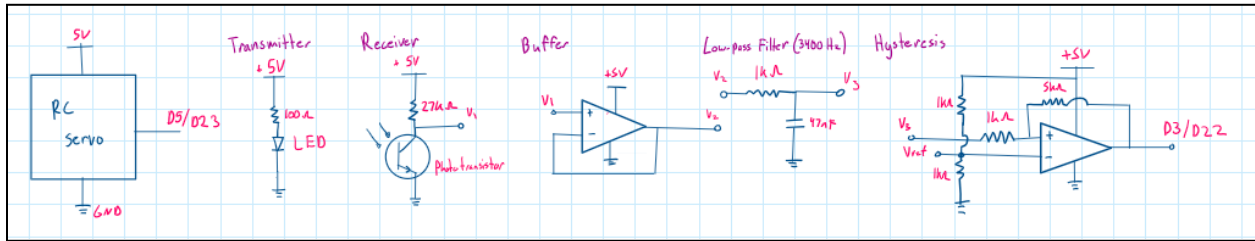
Jellyfish (DC Motor)



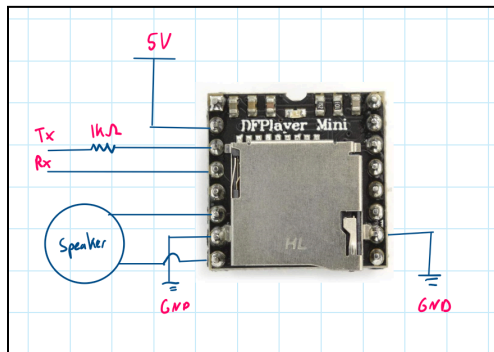
Flippers (Solenoid)



Gate and Krusty Krab (Servo+Optic)

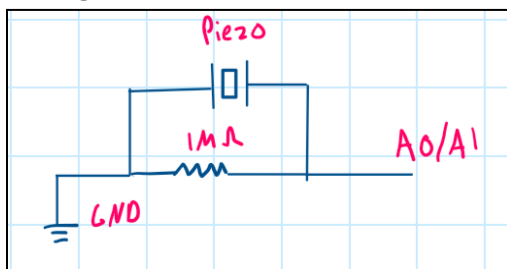


Speaker + Audio Module (datasheet: <https://m.media-amazon.com/images/I/C1LHhvsIk0S.pdf>)

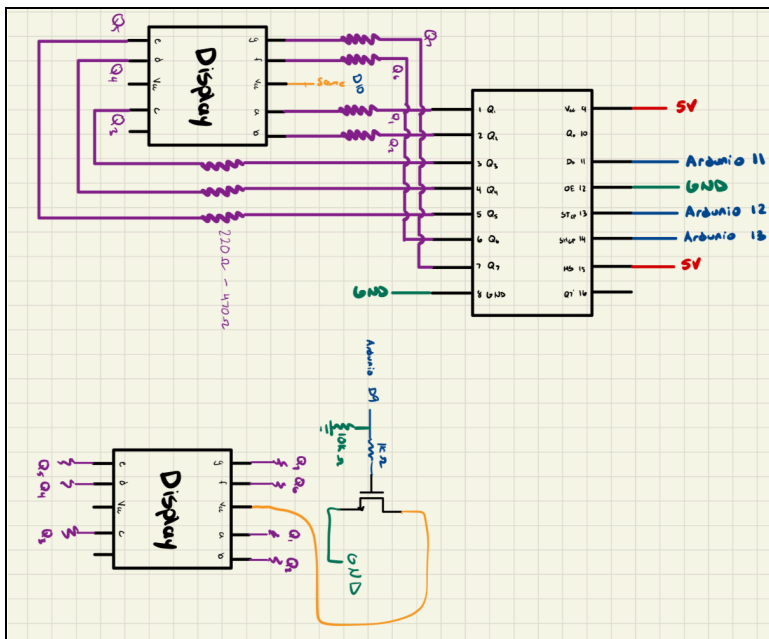


Tx = Pin 18, Rx = Pin 19

Spongebob/Squidward Home(Piezo)



Display



Pinball Machine Summary:

Our pinball machine was constructed primarily from plywood with several 3D printed components and electronics and hardware provided by the class. The playfield is slanted so the ball naturally rolls towards the losing region. A spring loaded plunger launches the ball onto the playfield, and two solenoid driven flippers allow the player to redirect the ball and keep it in play. Obstacles on the board include plywood walls, 3D printed SpongeBob themed houses, and plywood shapes of a jellyfish and Mr. Krab's arm are attached to an RC servo and DC motor. All components share a common ground with bypass capacitors across the power rails to stabilize supply. Comparing the CAD model with the final built machine highlights the importance of accurate modeling when translating a design into a physical system.

Points are earned through several sensor systems. Piezo sensors inside Squidward's and SpongeBob's houses detect impacts from the ball and produce voltage spikes that are read by an Arduino analog input. An IR emitter and receiver inside the Krusty Krab detect when the ball passes between them, awarding points and trigger Mr. Krab's arm (attached to RC servo) to move between positions. This detection circuit uses an op-amp buffer, low pass filter, and hysteresis comparator to reduce noise and prevent false triggers.

Another IR emitter and receiver pair detects when the ball reaches the losing region, allowing the Arduino to track rounds and reset the game when the ball is reloaded and both flipper buttons are pressed. The DC motor and solenoids are controlled using power MOSFETs so the Arduino can safely switch higher current loads. A pull down connection on the MOSFET gate ensures the device turns off if the Arduino disconnects, and flyback diodes across the motor and solenoids suppress inductive voltage spikes. The solenoid circuit also includes a capacitor and resistor to control activation and safely discharge stored energy.

A DFPlayer Mini audio module plays SpongeBob themed sounds during gameplay, and two seven segment displays are multiplexed using MOSFET switching to show the score. All hardware interfaces with the Arduino, which coordinates sensor inputs and actuator outputs. The block diagram illustrates the connections between components, while the FSM diagram represents the high level game states and transitions that control gameplay.

3. Safety, Ethics, and Diversity:

Since we are working with circuits for this project, we ensured that nothing will catch fire from unsafe operation (overheating) or shorting. We made sure to always check the secureness and connection of the wires including a common ground reference point. Multiple wires were connected between subsystem grounds for redundancy. Since the pinball machine involves launching a ball at high speed from a plunger, we implemented several safety measures to prevent potential injuries. Springs were added on both ends of the plunger to better control the launch speed of the ball, and the walls of the machine were built high enough to prevent stray balls from leaving the playfield. We also sanded down any chipped or sharp edges on the machine to reduce the risk of splinters and make it safer for people to handle. We also respected the ECE Makerspace through proper usage of all equipment and following rules for the space.

On ethics, we were mindful of minimizing waste and responsibly using the shared class resources. We returned unused components so that other/future students could use them. We also tested our system carefully before demonstrations to avoid damaging equipment or creating unsafe conditions in the makerspace.

For diversity, our theme is Spongebob, a cartoon that is recognized around the world and familiar to many people. We hope that this theme helps make the project engaging for a broad audience. We also tried to design the machine so that the gameplay is simple and easy to understand, allowing people from different backgrounds or technical experience levels to interact with it.